

Yu-Yun Chang

The Graduate Center, CUNY · 365 Fifth Avenue, New York, NY 10016
+1 (929) 217-7923 · ychang2@gc.cuny.edu · yuyunchang.github.io · [LinkedIn](#)

EDUCATION

Ph.D. in Economics , The Graduate Center, CUNY, New York, NY	Expected 2027
M.S. in Financial Engineering , NYU Tandon School of Engineering, New York, NY	2018
M.A. in International Economics , National Chung Cheng University, Taiwan	2015
B.B.A. in Economics , National Central University, Taiwan	2011

RESEARCH FIELDS

International Macroeconomics · Financial Econometrics · Time-Series Econometrics

JOB MARKET PAPER

Is the Dollar Reconnect Forecastable? Information Timing and Exchange-Rate Predictability

A growing literature shows that the U.S. dollar has reconnected with global financial conditions, reviving the view that exchange rates are tied to the macro-financial environment. I ask whether this reconnect makes exchange rates easier to forecast. Across thirteen currencies against the U.S. dollar, I compare forecasts built from equity information available at the forecast origin with a benchmark that also admits equity returns realized during the forecast window, holding the target and model fixed so that only the information set changes. The reconnect is large once realized returns are admitted, but it delivers little reliable forecasting gain at the origin: even after the global financial crisis, forecast-origin performance does not beat a random walk on average. The dollar is reconnected in realized data but not in monthly forecasts. A pre-registered extension measures how quickly early-month equity information is priced into exchange rates.

WORKING PAPERS

Exchange-Rate Adjustment and Reconnection to Fundamentals

- Studies whether long-run exchange-rate adjustment to macroeconomic fundamentals can be identified reliably when integration order, structural breaks, and cross-sectional dependence are treated as model-selection inputs rather than maintained assumptions.
- Builds a harmonized monthly panel of thirteen currencies against the U.S. dollar, together with interest differentials, relative prices, and relative industrial production, with currency-specific effective samples and documented data sources.
- Applies a staged specification framework spanning Johansen/VECM, ARDL-ECM, and no-anchor cases, with an analytical detectability benchmark that separates weak economic adjustment from limits imposed by sample length and statistical resolution.

Pooling Across Currencies: The Unit of Analysis in Exchange-Rate Models

- Asks whether exchange-rate relationships that are difficult to detect currency by currency become identifiable when information is pooled across thirteen currencies against the U.S. dollar, completing a three-part study of measurement limits: sample length, information timing, and unit of analysis.
- Structural part: re-examines long-run adjustment to macroeconomic fundamentals with panel error-correction estimators that allow heterogeneous short-run dynamics and cross-sectional dependence, and compares pooled detection with single-currency detectability benchmarks.
- Predictive part: compares currency-specific, fully pooled, and partially pooled forecasting models under identical forecast-origin information sets and out-of-sample evaluation, separating gains from pooling from common exposure to the dollar factor.

PUBLICATIONS AND PRESENTATIONS

Hung, J. and **Y.-Y. Chang** (2019). “Why Has the United States Continued to Receive Net Investment Income as a Debtor Country?” *The World Economy*, 42(3), 936–958.

Economics Brown Bag Seminar, The Graduate Center, CUNY	2025, 2026
Dissertation Seminar, The Graduate Center, CUNY	2024, 2025
Western Economic Association International Annual Conference, San Diego	2017

RESEARCH IN PROGRESS

Time-Varying Adjustment and the Exchange-Rate Disconnect

Develops a time-varying framework for exchange-rate adjustment across USD currency pairs, with TVP-VECM specifications where cointegration is supported and TVP-VAR specifications otherwise, to ask whether the disconnect is a stable feature of the data or varies across monetary, macroeconomic, and financial environments.

Regime-Dependent Exchange-Rate Predictability and Machine Learning

Studies whether the predictive content of macroeconomic and financial variables changes across low- and high-risk states defined by global market volatility, combining regime-specific gradient-boosting forecasts with SHAP attribution to compare the contributions of fundamentals, valuation signals, and global-risk indicators.

Decomposition-Based Machine Learning for Daily FX Forecasting

Develops a rolling daily-FX forecasting framework based on variational mode decomposition that assigns different models (ARIMA, recurrent neural networks, fuzzy-rule systems) to frequency components, with leakage-free rolling decomposition and genuinely out-of-sample evaluation.

TEACHING EXPERIENCE

Hunter College, City University of New York – Graduate Teaching Fellow/Assistant	2020–2024
Techniques of Economic Analysis	Fall 2020–2023
Microeconomic Theory	Fall 2020–2023
Introductory Econometrics	Fall 2020
Nonlinear Econometric Analysis	Spring 2021
Behavioral Economics	Spring 2022
International Trade	Spring 2021, 2023, 2024

APPLIED QUANTITATIVE AND MODEL VALIDATION PROJECTS

Independent Validation of a GARCH Volatility Model

Independent validation of a GARCH(1,1) model that supports Value-at-Risk estimation. Assessed conceptual soundness and parameter stability, replicated the model from first principles against a reference implementation, and benchmarked it against EWMA and asymmetric-GARCH challengers. Backtested VaR under Kupiec, Christoffersen, and Basel traffic-light criteria, and documented findings and remediation recommendations.

Point-in-Time Financial Data Pipeline

Built a release-date-aware data pipeline on FRED/ALFRED vintages, with point-in-time joins that restrict each observation to data available on its decision date, schema validation, publication-lag profiling, and automated data-quality reporting. Quantified the backtest gap between final-revised and point-in-time data. Deployed on AWS (S3, RDS, Lambda, EventBridge) under least-privilege IAM.

Validation of a Retrieval-Augmented LLM System

Validation framework for a retrieval-augmented generation system over regulatory guidance documents, built on a hand-verified evaluation set spanning in-scope, out-of-scope, and adversarial cases. Measured retrieval recall, answer faithfulness, and hallucination rates; ran prompt-sensitivity and non-determinism stability tests; and produced a failure taxonomy, a model card, and a validation report structured along SR 11-7 guidance. Built on Amazon Bedrock with S3 Vectors.

Model Monitoring and Inventory Framework

Monitoring layer that tracks models through a tiered inventory with materiality-based validation cycles. Detects input drift (population stability index, Kolmogorov–Smirnov), tracks performance decay against control limits, segments performance by volatility regime, and drives three-tier alerting with a remediation log. Reporting is automated through AWS Lambda, CloudWatch, and SNS.

TECHNICAL SKILLS

Econometrics: cointegration (Johansen, ARDL-ECM); VAR/VECM; time-varying parameter and state-space models; GARCH-family models; panel methods; structural breaks; bootstrap and multiple-testing inference; forecast evaluation.

Machine learning: gradient boosting; random forests; LASSO and ridge; neural networks; SHAP attribution; walk-forward validation.

Research practice: point-in-time data construction; provenance audits; pre-registered analysis protocols; Monte Carlo power analysis; reproducible pipelines.

Programming and data: Python (pandas, scikit-learn, statsmodels, XGBoost, arch); R; Stata; SQL and PostgreSQL; Git; L^AT_EX; Bloomberg; FRED/ALFRED; OECD and BIS databases.

CERTIFICATIONS, AWARDS, AND LANGUAGES

Certifications: Level II Candidate in the CFA Program · AWS Generative AI and AI Agents with Amazon Bedrock · IBM Building AI Agents and Agentic Workflows

Awards: Graduate Center Fellowship, CUNY (2019) · Graduate Scholarship, NYU Tandon (2016) · Graduate Scholarship, National Chung Cheng University (2014)

Languages: Mandarin Chinese (native) · English (fluent)

REFERENCES

Merih Uctum (Advisor)

Professor of Economics

The Graduate Center, CUNY

muctum@gc.cuny.edu

Juann Hung

Professor of Economics

Xi'an Jiaotong-Liverpool University

Juann.Hung@xjtlu.edu.cn